Analytic soundness of the optional coefficient audit

Let $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, with $-R_- \le X \le R_+$. Write $\rho = \mathbb{E}|X|^3$, $\gamma = \mathbb{E}X^3$, $\beta = \mathbb{E}[X|X|]$, $K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2$. For the actual sign split let $V = \mathbb{E}[X \mid \mathbf{1}_{\{X>0\}}]$, $W = X - V$ and $\theta_\pm = \mathbb{E}[W^2; \pm X > 0]$. The signed conditional and central-gap clauses below assume an already proved gap $(-A, B)$ with $A, B > 0$; the moment matrices also allow an atom at zero. Conditional centering proves, for signwise constant h and $j \ge 2$,

$$\mathbb{E}[h(|X|^j - |V|^j)] = \mathbb{E}\left[hW^2 \sum_{k=0}^{j-2} (k+1)|V|^k |X|^{j-2-k}\right].$$

Indeed multiply the quotient by $(|X|-|V|)^2$ and restore $j|V|^{j-1}(|X|-|V|)$, whose conditional expectation is zero. For every support gap (u, v) put $\tau_{u,v} = \mathbb{E}[X; X > t]$ at $t \in (u, v)$, and define

$$J_1 = \sum \tau_{u,v} \int_u^v (t-u)t\,dt, \qquad J_2 = \sum \tau_{u,v} \int_u^v (t-u)t|t|\,dt.$$

Use subscripts c, o for the central and outer gap contributions. Let $m = \mathbb{E}X_+ = \mathbb{E}X_-$, $T_+ = \mathbb{E}X_+^3 - mB^2$ and $T_- = \mathbb{E}X_-^3 - mA^2$. The common nonnegative outer budget $\mathcal{B}$ appearing below is an explicit input hypothesis, not acquired by polynomial acceptance. Here $\sigma = \sqrt{10}$; any proved positive lower prices may replace $\sigma - 1, \sigma + 1$ in a weaker budget.

Why the coefficient formulas bound one marginal. There are three underlying calculations. They use the same signwise moments, conditional variances and gap budget throughout; the exact coefficient object records their polynomial expansions.

First, on either sign let $Y = |X| \in [0, r]$ and write $s_j = \mathbb{E}[Y^j; \text{that sign}]$. Both signs have $s_1 = m$. For $H_j = \left(\begin{smallmatrix} s_j & s_{j+1} \\ s_{j+1} & s_{j+2} \end{smallmatrix}\right)$,

$$H_j \succeq 0 \quad (0 \le j \le 3), \qquad rH_j - H_{j+1} \succeq 0 \quad (0 \le j \le 2).$$

Their quadratic forms integrate, respectively, $Y^j(a+bY)^2$ and $Y^j(r-Y)(a+bY)^2$. These identities prove the whole moment and support-contraction family, separately at the two support radii. A quotient deduction from $\left(\begin{smallmatrix} A & B \\ B & C \end{smallmatrix}\right) \succeq 0$ uses $C \ge B^2/A$ only when $A > 0$. At $A = 0$ it uses $B = 0$, $C \ge 0$; an interval meeting zero may simply discard the nonnegative correction. It may not divide by zero or square an unlicensed signed endpoint.

Second, the conditional Taylor identity displayed above integrates its polynomial remainder against the nonnegative measures $d\nu_\pm = (X - V)^2 \mathbf{1}_\pm dP$, of masses $\theta_\pm$. With $\bar{a}_\pm = |\mathbb{E}[X \mid \pm]|$, $\gamma_V = \mathbb{E}V^3$, it gives

$$K_1(X) - K_1(V) = \int F(x, \bar{a}_+; \rho + \gamma_V)\, d\nu_+ - \int F(x, \bar{a}_-; \rho - \gamma_V)\, d\nu_-,$$
$$F(x, a; t) = x^2 + 2ax + 3a^2 - 3 - \tfrac{3}{2}t(x + 2a).$$

Indeed expand the product as $\rho\gamma - \rho_V\gamma_V = \rho(\gamma - \gamma_V) + \gamma_V(\rho - \rho_V)$ and use the conditional identities at orders two, three and four. The factored derivatives certify $F_t \le 0$ and the positive Hessian $2(dx + da)^2 + 4(da)^2$. Thus a rectangle maximum is at a corner; a minimum is among its corners, its clipped edge stationary points and its interior stationary point. All are included in the exact coefficient map. If the two signed integrands lie in $[L, U]$, their sum lies in $[\Theta \min(0, L), \Theta \max(0, U)]$ when $\theta_+ + \theta_- \le \Theta$. This includes zero noise without division.

Third, direct integration over the central gap $(-A, B)$ gives

$$J_{1,c} = mP_1(A, B), \quad J_{2,c} = mP_2(A, B),$$
$$P_1 = B^3/3 + AB^2/2 - A^3/6, \qquad P_2 = B^4/4 + AB^3/3 - A^4/12.$$

Both polynomials increase in B and their A derivatives change sign only at $A = B$. Their extrema therefore use the indicated endpoints and the clipped point $A = B$. For $G = \rho P_1 - P_2$, the complete central bound uses the degree-four tensor Bernstein coefficients on the closed (A, B) rectangle. The basis is nonnegative and sums to one, so retain both m_l and m_u times every coefficient. This handles negative coefficients and zero-width rectangles. The result is a maximum of affine functions of ρ, hence convex. The checked inverse basis identities prove coverage of the entire rectangle, not only a grid.

For a positive outer gap (a, b), $a \geq B$, the square inequality $(t - a)(\rho - t) \leq (\rho - B)_+^2/4$ and signwise Tonelli give

$$3(\rho J_1 - J_2)_o \leq \tfrac{3}{8}\{(\rho - B)_+^2 T_+ + (R_- - \rho)_+^2 T_-\}, \qquad -R_- T_-/2 \leq J_{1,o} \leq R_+ T_+/2.$$

For the first bound on a negative gap (u, v), put $x = -t$ and $b = -u$: its integrand is $(b - x)x(x - \rho)$, whose positive part is at most $x(R_- - \rho)_+^2/4$. Enlarge each union of gaps to its complete outer ray; the integrand is nonnegative, and Tonelli gives $\int_B^\infty t\tau(t)\,dt = T_+/2$ and its negative counterpart. This proves the factor $3/8$ even if the support has positive Lebesgue measure. For the second bound, $0 \leq t - u \leq R_+$ on positive outer gaps and $0 \leq t - u \leq R_-$ on negative outer gaps; integrate $|t|\tau(t)$ over the complete corresponding ray. Endpoint atoms do not change these Lebesgue integrals.

These costs share the one budget $(\sigma - 1)T_+ + (\sigma + 1)T_- \leq \mathcal{B}$ as a separate premise. Intersect it with any proved individual caps before pricing a linear cost. Without caps, use

$$b_+ T_+ + b_- T_- \leq (\mathcal{B}) \max\{0, b_+/(\sigma - 1), b_-/(\sigma + 1)\}.$$

The zero option is necessary for signed costs. Sum costs before this single pricing step; intersect alternative upper bounds afterwards. Every coefficient interval is a necessary condition on this one law. The analytic formation of the central gap and the prior support bounds remain prerequisites; coefficient acceptance does not establish them.